Description of Additional Supplementary Files

File Name: Supplementary Data 1

Description: Assessment key parameters. Tables with key parameters. **Sheet 1**) QC of PGT-M and PGT-SR samples at 10X coverage. **Sheet 2**) QC of sub-sampled samples from family 1 and family 2. **Sheet 3**) QC of PGT-MT samples. PGT: preimplantation genetic testing; M: monogenic; SR: structural rearrangement; MT: mitochondrial DNA disorder; QC: quality control.

File Name: Supplementary Data 2

Description: Number of informative SNPs at ROI. The number of informative SNPs 2 Mb up- and downstream the ROI for each genetic indication and the concordance with affected or unaffected maternal/paternal haplotypes. GBS-PGT (**sheet1**), WGS-PGT (**sheet2**). WGS: whole genome sequencing; GBS: genotyping-by-sequencing; ROI: region of interest; SNP: single nucleotide polymorphism, PGT: preimplantation genetic testing.

File Name: Supplementary Data 3

Description: SNV detection in families from clinically implemented diagnostic WGS-PGT. Overview of the genetic variants in families that underwent clinical WGS-PGT. Each row corresponds to a unique family with a genetic variant, the mode of inheritance, the affected proband, the paternal and/or maternal variant, the type of variant and whether the variant could be directly detected. OMIM: Online Mendelian Inheritance in Man; X-R: X-linked recessive; X-D: X-linked dominant; AR: autosomal recessive; AD: autosomal dominant.

File Name: Supplementary Data 4

Description: Tables with breakpoint pairs identified by Manta in the carrier parent and embryos of families with translocations. Karyotyping results, a schematic of the translocation and breakpoints estimated from VeriSeq PGT-SR are also included for each family. **Sheet1**) Family 1, **sheet2**) family 19, **sheet3**) family 9, **sheet 4**) Copy Number Plots. From top to bottom, paternal haplarithms, maternal haplarithms, logR profile from QDNAseq and copy number profiles obtained from VeriSeq. In the paternal haplarithm and maternal haplarithm, each point indicates the parentally informative SNPs in blue (M1/P1) and red (P2/M2). The segmented parentally informative SNPs are shown as dark blue (M1/P1) or dark red (M2/P2). In the copy number profile from QDNAseq each point indicates the logR value (log2 copy number ratio between observed and expected) with a bin size of 100 kb and the segmented logR is depicted in orange. In the VeriSeq track, each point represents the copy number value of a bin sized ~1Mb. The green and red lines indicate the threshold for detection of copy number gain or loss, respectively. PGT: preimplantation genetic testing: SR: structural rearrangements; SNP: single nucleotide polymorphism.